Supplementary Material

Appendix A
Proof of Lemma 1

Lemma 1 (Minimum Queue Length)**.** *At time t, the minimum bottleneck queue length can be given by*

$$q_{\min}(t) = \sum_{i=1}^{N} w_i(t) - \sum_{i=1}^{N} B_i(t) - Cd \tag{1}$$

Proof. Since the arrival traffic at bottleneck queue exhibits ON/OFF pattern, the queue length reaches its minimum value when arrival traffic from all sources is at the end of OFF period. Specifically, the batch buffer has delivered a batch of packets to the send buffer and the first packet in this batch is just arriving at the queue. In other words, it's d_1 after batch buffer delivers the batch to the send buffer. On the other hand, the number of packets sent out by transport layer is $\sum_i w_i(t)$. These packets are distributed in: ① batch buffer, ② send buffer, ③ flying between sender and bottleneck queue, ④ bottleneck queue, ⑤ flying between bottleneck queue and receiver, ⑥ arrived at receiver but the sender has not received their acknowledgements. Since $\sum_i \frac{w_i(t)}{R(t)} = C$, the number of packets in ① is $\sum_i b_i(t) = Cd_1$. The total number of packets in ② and ③ is $\sum_i B_i(t)$. The number of packets in ④, ⑤ and ⑥ is $q(t)$, Cd_2, and $C(d_1 + d_2)$, respectively. Combining $d = 2(d_1 + d_2)$, we have

$$\sum_{i=1}^{N} w_i(t) = q_{\min}(t) + \sum_{i=1}^{N} B_i(t) + Cd \tag{2}$$

Which is equivalent to equation (1). ■

Appendix B
Proof of Lemma 2

Lemma 2 (Maximum Queue Length)**.** *At time t, the maximum bottleneck queue length can be given by*

$$q_{\max}(t) = \sum_{i=1}^{N} w_i(t) - \frac{\sum_{i=1}^{N} [B_i(t-\tau_i) w_i(t)]}{CR(t)} - Cd \tag{3}$$

where $\tau_i = \frac{B_i(t-\tau_i)}{C} + d_1$.

Proof. Similarly, since the arrival traffic at bottleneck queue exhibits ON/OFF pattern, the queue length reaches its maximum value when arrival traffic from all sources is at the end of ON period. This moment is d_1 after the send buffer sent out all packets. During this time, the evolution of $s_i(t)$ is shown in Fig. 1. Therefore, the number of accumulated packets in batch buffer i can be given by $\frac{w_i(t)}{R(t)} \cdot \left[\frac{B_i(t-\tau_i)}{C} + d_1\right]$. The total number of packets in batch buffer can be given by $\frac{\sum_{i=1}^{N}[B_i(t-\tau) w_i(t)]}{CR(t)} + Cd_1$. On the other hand, the total number of packets in other places can be given by $q(t) + 2Cd_2 + Cd_1$. Thus, we have

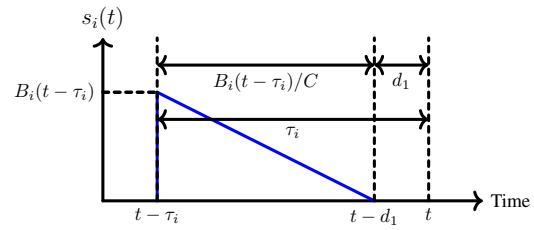


Fig. 1. The number of packets in send buffer before the bottleneck queue length reaches its maximum value

$$\sum_{i=1}^{N} w_i(t) = \frac{\sum_{i=1}^{N} [B_i(t-\tau) w_i(t)]}{CR(t)} + q_{\max}(t) + Cd \tag{4}$$

Which is equivalent to equation (3). ■

Appendix C
Proof of Theorem 1

Theorem 1. *With ECN*, if the window size when packets are marked is $w(t)$, and it takes Δt for the senders to receive congestion signal after packets are marked with ECN. Then the amplitude of queue length oscillations can be given by*

$$A = \frac{N}{2} w(t) + NB(t + \Delta t) - B(t) \tag{5}$$

Proof. In ECN*, the TCP sender will cut its window in half when receiving an ECN feedback.. If the window size when packets are marked is $w(t)$, then $w(t+\Delta t) = \frac{w(t)}{2}$. Therefore, the amplitude of queue length oscillations can be given by

$$\begin{aligned} A &= q_{\max}(t) - q_{\min}(t + \Delta t) \\ &= \frac{N}{2} w(t) + NB(t + \Delta t) - B(t) \end{aligned} \tag{6}$$

■

Appendix D
Proof of Corollary 1

Corollary 1. *With ECN*, when LSO is enabled, the amplitude of queue length oscillations can be given by*

$$\begin{cases} \frac{[N(\beta+1)-2\beta](Cd+K)}{2(N-\beta)} \leqslant A \leqslant \frac{[N(\beta+1)-2\beta](Cd+K)}{2N(1-\beta)}, \\ \qquad \textit{if } \beta w(t) < L \\ \frac{(\beta+1)(Cd+K+L)}{2} - L \leqslant A \leqslant \frac{(\beta+1)(Cd+K)+(N\beta+N-2)L}{2}, \\ \qquad \textit{if } L \leqslant \beta w(t) < 2L \\ \frac{Cd+K}{2} + (N - \frac{1}{2})L \leqslant A \leqslant \frac{Cd+K}{2} + (\frac{3}{2}N - 1)L, \\ \qquad \textit{if } \beta w(t) \geqslant 2L \end{cases} \tag{7}$$

Proof. When LSO is enabled, the batching size $B(t)$ and $B(t+\Delta t)$ can be given by

$$\begin{cases} B(t)=\beta w(t), & B(t+\Delta t)=\frac{\beta}{2}w(t), & \text{if } \beta w(t)<L \\ B(t)=L, & B(t+\Delta t)=\frac{\beta}{2}w(t), & \text{if } L\leqslant \beta w(t)<2L \\ B(t)=L, & B(t+\Delta t)=L, & \text{if } \beta w(t)\geqslant 2L \end{cases} \tag{8}$$

Substituting them into equation (5), we get

$$A=\begin{cases} \frac{N(\beta+1)-2\beta}{2}w(t), & \text{if } \beta w(t)<L \\ \frac{N(\beta+1)}{2}w(t)-L, & \text{if } L\leqslant \beta w(t)<2L \\ \frac{N}{2}w(t)+(N-1)L, & \text{if } \beta w(t)\geqslant 2L \end{cases} \tag{9}$$

On the other hand, $q_{\min}\leqslant K\leqslant q_{\max}$, from which we have

$$\begin{cases} \frac{Cd+K}{N-\beta}\leqslant w(t)\leqslant \frac{Cd+K}{N(1-\beta)}, & \text{if } \beta w(t)<L \\ \frac{Cd+K+L}{N}\leqslant w(t)\leqslant \frac{Cd+K}{N}+L, & \text{if } \beta w(t)\geqslant L \end{cases} \tag{10}$$

Substituting inequality (10) into equation (9), we can get inequality (7). ■

Appendix E
Proof of Corollary 2

Corollary 2. *With ECN*, when LSO is enabled, to avoid throughput loss, the ECN threshold K should meet the following inequality*

$$K\geqslant \begin{cases} \frac{(1+\beta)N-2\beta}{(1-\beta)N}Cd, & \text{if } \beta w(t)<2L \\ Cd+(2N-1)L, & \text{if } \beta w(t)\geqslant 2L \end{cases} \tag{11}$$

Proof. To avoid throughput loss, the queue should not underflow, namely,

$$q_{\min}(t+\Delta t)\geqslant 0 \tag{12}$$

On the other hand,

$$q_{\min}(t+\Delta t)=\begin{cases} \frac{N}{2}(1-\beta)w(t)-Cd, & \text{if } \beta w(t)<2L \\ \frac{N}{2}w(t)-NL-Cd, & \text{if } \beta w(t)\geqslant 2L \end{cases} \tag{13}$$

Combining equation (10), we must have

$$\begin{cases} \frac{N}{2}(1-\beta)\frac{Cd+K}{N-\beta}\geqslant Cd, & \text{if } \beta w(t)<2L \\ \frac{Cd+K+L}{2}\geqslant NL+Cd, & \text{if } \beta w(t)\geqslant 2L \end{cases} \tag{14}$$

This inequality is equivalent to inequality (11). ■

Appendix F
Proof of Theorem 2

Theorem 2. *With DCTCP, if the window size when packets are marked is $w(t)$, and it takes Δt for the senders to receive congestion signal after packets are marked with ECN. Then the amplitude of queue length oscillations can be given by*

$$A=\frac{\alpha}{2}Nw(t)+NB(t+\Delta t)-B(t) \tag{15}$$

Proof. Different than ECN*, with DCTCP, the sender will update its window as $w=w\times(1-\frac{\alpha}{2})$ when receiving an ACK with ECE bit, where α denotes the fraction of marked packets. Therefore, if the window size when packets are marked is $w(t)$, and the sender receives congestion signal after Δt, then we have $w(t+\Delta t)=(1-\frac{1}{\alpha})w(t)$. The amplitude of queue length oscillations can be given by

$$\begin{aligned} A&=q_{\max}(t)-q_{\min}(t+\Delta t) \\ &=\frac{\alpha}{2}Nw(t)+NB(t+\Delta t)-B(t) \end{aligned} \tag{16}$$

■

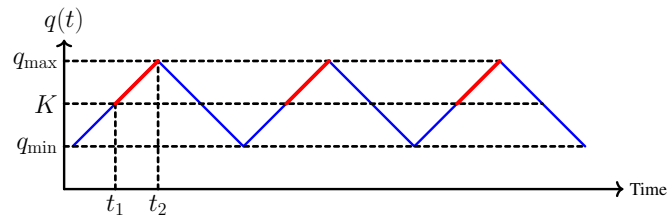


Fig. 2. Queue length evolution of DCTCP with batching

Appendix G
Proof of Corollary 3

Corollary 3. *With DCTCP, when LSO is enabled, there are three cases:*

(1) If $\beta w(t)<L$, the amplitude of queue length oscillations can be given by

$$\begin{aligned} &\tfrac{(N-1)\beta}{N-\beta}(Cd+K)\leqslant A\leqslant \\ &\quad \tfrac{(N-1)\beta}{N(1-\beta)}(Cd+K)+\sqrt{\tfrac{N(1-\beta)(Cd+K)}{2}} \end{aligned} \tag{17}$$

(2) If $L\leqslant \beta w(t)<\frac{L}{1-\sqrt{\frac{Nw(t)-L-Cd-K}{2(N-1)Lw(t)}}}$, the amplitude of queue length oscillations can be given by

$$\begin{aligned} &\beta(Cd+K)-(1-\beta)L\leqslant A\leqslant \\ &\quad N(1-\beta)\sqrt{\tfrac{N}{2(Cd+K+NL)}}+\beta(Cd+K)+(\beta N-1)L \end{aligned} \tag{18}$$

(3) If $\beta w(t)\geqslant \frac{L}{1-\sqrt{\frac{Nw(t)-L-Cd-K}{2(N-1)Lw(t)}}}$, the amplitude of queue length oscillations can be given by

$$(N-1)L\leqslant A\leqslant \sqrt{\tfrac{N(Cd+K+NL)}{2}}+(N-1)L \tag{19}$$

Proof. When LSO is enabled, the batch size $B(t)$ and $B(t+\Delta t)$ can be given by

$$\begin{cases} B(t)=\beta w(t), B(t+\Delta t)=(1-\frac{\alpha}{2})\beta w(t), \text{ if } \beta w(t)<L \\ B(t)=L, B(t+\Delta t)=(1-\frac{\alpha}{2})\beta w(t), \text{if } L\leqslant \beta w(t)<\frac{2L}{2-\alpha} \\ B(t)=L, B(t+\Delta t)=L, \text{ if } \beta w(t)\geqslant \frac{2L}{2-\alpha} \end{cases} \tag{20}$$

Substituting them into equation (15), we have

$$A=\begin{cases} \left[\frac{\alpha}{2}N(1-\beta)+(N-1)\beta\right]w(t), & \text{if } \beta w(t)<L \\ \left[\frac{\alpha}{2}N(1-\beta)+N\beta\right]w(t)-L, & \text{if } L\leqslant \beta w(t)<\frac{2L}{2-\alpha} \\ \frac{\alpha}{2}Nw(t)+(N-1)L, & \text{if } \beta w(t)\geqslant \frac{2L}{2-\alpha} \end{cases} \tag{21}$$

To further expand this equation, we need to estimate α. The queue length evolution when packets are marked is shown in Fig. 2. Among a single batch of packets, packets arriving between t_1 and t_2 are marked with ECN. In $[t_1,t_2]$, the queue length is increasing at a rate of $(N-1)C$, thus $t_2-t_1=\frac{q_{max}-K}{(N-1)C}$. On the other hand, during $[t_1,t_2]$, each sender is sending packets at a rate of C. Therefore, the number of marked packets during this time is $\frac{q_{max}-K}{N-1}$. The sender will receive congestion signal about an RTT later. During an RTT, there are $1/\beta$ batches if $\beta w(t)<L$ and $w(t)/L$ batches if $\beta w(t)\geqslant L$. Therefore, the number of marked packets in each flow can be given by $\frac{(q_{max}-K)}{(N-1)\beta}$ if $\beta w(t)<L$ and $\frac{(q_{max}-K)w(t)}{(N-1)L}$ if $\beta w(t)\geqslant L$.

On the other hand, let $S(W_1,W_2)$ denote the number of sent packets when the window of sender increases from W_1

to W_2, then we have $S(W_1, W_2) = \frac{W_2^2 - W_1^2}{2}$. Thus, we can estimate α by

$$\alpha = \begin{cases} \frac{\frac{q_{max}-K}{(N-1)\beta}}{S\left(w(t)(1-\frac{\alpha}{2}), w(t)\right)}, & \text{if } \beta w(t) < L \\ \frac{\frac{(q_{max}-K)w(t)}{(N-1)L}}{S\left(w(t)(1-\frac{\alpha}{2}), w(t)\right)}, & \text{if } \beta w(t) \geqslant L \end{cases} \tag{22}$$

As $q_{\max}(t) = Nw(t) - B(t) - Cd$, we have

$$\alpha^2 - \frac{1}{4}\alpha^3 = \begin{cases} \frac{2(Nw(t)-\beta w(t)-Cd-K)}{(N-1)\beta w^2(t)}, & \text{if } \beta w(t) < L \\ \frac{2(Nw(t)-L-Cd-K)}{(N-1)Lw(t)}, & \text{if } \beta w(t) \geqslant L \end{cases} \tag{23}$$

As α is usually very small, this equation can be simplified as

$$\alpha \approx \begin{cases} \frac{1}{w(t)}\sqrt{\frac{2(Nw(t)-\beta w(t)-Cd-K)}{(N-1)\beta}}, & \text{if } \beta w(t) < L \\ \sqrt{\frac{2(Nw(t)-L-Cd-K)}{(N-1)Lw(t)}}, & \text{if } \beta w(t) \geqslant L \end{cases} \tag{24}$$

In the next, we analyze the amplitude case by case.

(1) $\beta w(t) < L$

Plugging equation (24) into equation (21), we get

$$A = N(1-\beta)\sqrt{\frac{(N-\beta)w(t)-(Cd+K)}{2(N-1)\beta}} + (N-1)\beta w(t) \tag{25}$$

Finally, plugging (10) into it inequality (17).

(2) $L \leqslant \beta w(t) < \frac{2L}{2-\alpha}$

Plugging equation (24) into equation (21), we get

$$A = N(1-\beta)\sqrt{\frac{Nw(t)-(Cd+K)-L}{2(N-1)Lw(t)}} + N\beta w(t) - L \tag{26}$$

Note that A increases monotonically with the increasing of $w(t)$ when $\frac{Cd+K+L}{N} \leqslant w(t) \leqslant \frac{Cd+K}{N} + L$. Therefore, combining (10), we have inequality (18).

(3) $\beta w(t) \geqslant \frac{2L}{2-\alpha}$

Plugging equation (24) into equation (21), we get

$$A = N\sqrt{\frac{[Nw(t)-(Cd+K)-L]w(t)}{2(N-1)L}} + (N-1)L \tag{27}$$

A increases monotonically with the increasing of $w(t)$ when $\frac{Cd+K+L}{N} \leqslant w(t) \leqslant \frac{Cd+K}{N} + L$. Therefore, combining (10), we have equation (19). ■

Appendix H
Proof of Corollary 4

Corollary 4. *With DCTCP, when LSO is enabled, there are three cases:*

(1) If $\beta w(t) < L$, to avoid throughput loss, the ECN threshold (denoted by K) should satisfy the following inequalities

$$K \geqslant \begin{cases} \frac{(N-1)\beta}{N(1-\beta)}Cd + \frac{(N-\beta)^2}{8(N-1)\beta}, \\ \qquad \text{if } \left[\frac{(N-\beta)}{\beta(N-1)}\right]^2 \frac{N(1-\beta)}{8} \leqslant Cd + K \\ \frac{N(1-\beta)+\sqrt{N^2(1-\beta)^2+8NCd(1-\beta)}}{4}, \\ \qquad \text{if } \left[\frac{(N-\beta)}{\beta(N-1)}\right]^2 \frac{N(1-\beta)}{8} > Cd + K \end{cases} \tag{28}$$

(2) If $L \leqslant \beta w(t) < \frac{L}{1-\sqrt{\frac{Nw(t)-L-Cd-K}{2(N-1)Lw(t)}}}$, to avoid throughput loss, the ECN threshold should meet the following inequalities

$$K \geqslant \frac{\beta}{1-\beta}Cd - L \tag{29}$$

(3) If $\beta w(t) \geqslant \frac{L}{1-\sqrt{\frac{Nw(t)-L-Cd-K}{2(N-1)Lw(t)}}}$, to avoid throughput loss, the ECN threshold should meet the following inequalities

$$K \geqslant (N-1)L \tag{30}$$

Proof. To avoid throughput loss, the queue length should be always larger than 0, namely,

$$q_{\min}(t+\Delta t) = Nw(t+\Delta t) - NB(t+\Delta t) - Cd \geqslant 0 \tag{31}$$

There are three cases:

(1) $\beta w(t) < L$

In this case, $B(t+\Delta t) = (1-\frac{\alpha}{2})\beta w(t)$. Thus, inequality (31) can be rewritten as

$$w(t) - \sqrt{\frac{1}{2(N-1)\beta}} \cdot \sqrt{(N-\beta)w(t) - (Cd+K)} \geqslant \frac{Cd}{N(1-\beta)} \tag{32}$$

Let $f(w) = w(t) - \sqrt{\frac{1}{2(N-1)\beta}}\sqrt{(N-\beta)w(t) - (Cd+K)}$. Then $f(w)$ reaches its global minimum when

$$w = w^* = \frac{Cd+K}{N-\beta} + \frac{N-\beta}{8(N-1)\beta} \tag{33}$$

However, since $\frac{Cd+K}{N-\beta} \leqslant w(t) \leqslant \frac{Cd+K}{N(1-\beta)}$, there are two cases:

a) $\left[\frac{(N-\beta)}{\beta(N-1)}\right]^2 \frac{N(1-\beta)}{8} \leqslant Cd + K$

In this case, $\frac{Cd+K}{N-\beta} \leqslant w^* \leqslant \frac{Cd+K}{N(1-\beta)}$. Therefore, $f(w)$ reaches its minimum when $w = w^*$, namely,

$$f(w)_{\min} = f(w^*) = \frac{Cd+K}{N-\beta} - \frac{N-\beta}{8(N-1)\beta} \tag{34}$$

Inequality (32) can be rewritten as

$$\frac{Cd+K}{N-\beta} - \frac{N-\beta}{8(N-1)\beta} \geqslant \frac{Cd}{N(1-\beta)} \tag{35}$$

Which can be expressed as

$$K \geqslant \frac{(N-1)\beta}{N(1-\beta)}Cd + \frac{(N-\beta)^2}{8(N-1)\beta} \tag{36}$$

b) $\left[\frac{(N-\beta)}{\beta(N-1)}\right]^2 \frac{N(1-\beta)}{8} > Cd + K$

In this case, $w^* > \frac{Cd+K}{N(1-\beta)}$. Therefore,

$$f(w)_{\min} = f\left(\frac{Cd+K}{N(1-\beta)}\right) = \frac{Cd+K}{N(1-\beta)} - \sqrt{\frac{Cd+K}{2N(1-\beta)}} \tag{37}$$

Inequality (32) can be rewritten as

$$\frac{Cd+K}{N(1-\beta)} - \sqrt{\frac{Cd+K}{2N(1-\beta)}} \geqslant \frac{Cd}{N(1-\beta)} \tag{38}$$

Which gives

$$K \geqslant \frac{N(1-\beta) + \sqrt{N^2(1-\beta)^2 + 8NCd(1-\beta)}}{4} \tag{39}$$

(2) $L \leqslant \beta w(t) < \frac{2L}{2-\alpha}$

Combining (20) and (24), we can rewrite inequality (31) as

$$w(t) - \sqrt{\frac{(Nw(t) - L - Cd - K)w(t)}{2(N-1)L}} \geqslant \frac{Cd}{N(1-\beta)} \tag{40}$$

Let $f(w) = w - \frac{1}{\sqrt{2(N-1)L}}\sqrt{(Nw - L - Cd - K)w}$, then $f(w)$ reaches its minimum when

$$\sqrt{N - \frac{Cd + K + L}{w}} = \sqrt{2(N-1)L} - \sqrt{2(N-1)L - N} \quad (41)$$

Note that $2(N-1)L \gg N$. Therefore, $\sqrt{N - \frac{Cd+K+L}{w}} \approx 0$, which gives $w \approx \frac{Cd+K+L}{N}$. As a result, we can approximate the minimum of $f(w)$ as

$$f(w)_{\min} \approx \frac{Cd + K + L}{N} \quad (42)$$

Therefore, (40) can be rewritten as $\frac{Cd+K+L}{N} \geqslant \frac{Cd}{N(1-\beta)}$, which is equivalent to (29).

(3) $\beta w(t) \geqslant \frac{2L}{2-\alpha}$

In this case, we can rewrite inequality (31) as

$$w(t) - \sqrt{\frac{(Nw(t) - L - Cd - K)w(t)}{2(N-1)L}} \geqslant L + \frac{Cd}{N} \quad (43)$$

Let $f(w) = w - \frac{1}{\sqrt{2(N-1)L}}\sqrt{(Nw - L - Cd - K)w}$. As is analyzed previously, $f(w)_{\min} \approx \frac{Cd+K+L}{N}$. Therefore, (43) can be rewritten as $\frac{Cd+K+L}{N} \geqslant L + \frac{Cd}{N}$, which is equivalent to (30). ■

Appendix I
Performance of ECN marking Based on Average Queue Length

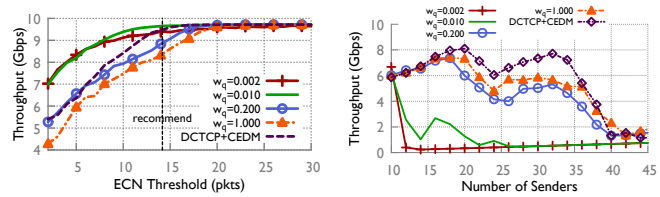
As is mentioned in §IV-A, if the switch marks packets based on the average queue length, the ECN mismarking can be mitigated. In this part, we use simulations to show that although average-queue-length-based ECN marking scheme can mitigate ECN mismarkings, it will reduce the responsiveness of end hosts to the fan-in congestion.

In the simulation, 50 hosts are connected to the same switch. The link capacity is 10Gbps and the base RTT is 100μs. The switch has 150KB buffer in each port. End hosts use DCTCP as their congestion control algorithms. Switch marks packets according to the average queue length, which is calculated by exponential weighted moving average (EWMA) as in RED:

$$avg_q = (1 - w_q)avg_q + w_q q \quad (44)$$

where avg_q denotes the average queue length, q denotes the instantaneous queue length, and w_q is the weight. Other settings are the same as those in previous simulations.

First, we evaluate the throughput when two senders are sending packets to the same receiver, sharing the same bottleneck. Fig.3(a) shows the throughput when $w_q = 0.002, 0.01, 0.2, 1.0$, respectively. The throughput is higher with smaller w_q. This is easy to understand since a larger transient increment of queue length can be accommodated with a smaller w_q. With a proper choice of w_q (i.e., $w_q = 0.01$), the throughput can be larger than DCTCP+CEDM. However, such a parameter setting can lead to poor performance in the incast scenario, in which senders should respond to congestion as soon as possible to avoid packet droppings. Specifically, as is shown in Fig.3(b), the throughput begins to collapse when the number of senders is only 10 when $w_q = 0.01$. In



(a) Overall throughput (b) Incast performance

Fig. 3. Performance when ECN is marked according to average queue length

conclusion, although average-queue-length-based ECN marking scheme can mitigate ECN mismarking, it can reduce the responsiveness of end hosts to fan-in congestion.